Unit-2

Lab experiments

Experiment 1: Smart Fitness Watch Product Description

Aim:

To design Zero-shot, One-shot, and Few-shot prompts to generate a product description for a Smart Fitness Watch.

Procedure:

1. Open Google Colab and create a new notebook.
2. Create three prompts using Zero-shot, One-shot, and Few-shot techniques.
3. Include product features and benefits in the prompts.
4. Display the prompts using Python's print() function.
5. Compare the prompts based on the instructions and examples provided.

Prompt:

```
zero_shot = """
Write a product description for a Smart Fitness Watch.
Include its features, benefits, design, and target customers.
"""

one_shot = """
Example:
Product: Wireless Earbuds
Description: Enjoy clear sound and wireless convenience.

Now write a similar product description for a Smart Fitness Watch.
Include its features and benefits.
"""

few_shot = """
Example 1:
Product: Smart Bottle
Description: Tracks water intake and reminds users to drink water.

Example 2:
Product: Fitness Band
```

Description: Tracks steps and daily activity.

Now write a product description for a Smart Fitness Watch.
Include heart-rate monitoring, sleep tracking, and notifications.
"""

print("Zero-shot Prompt:\n", zero_shot)
print("\nOne-shot Prompt:\n", one_shot)
print("\nFew-shot Prompt:\n", few_shot)

Result:

Three prompts were designed to generate a Smart Fitness Watch product description using Zero-shot, One-shot, and Few-shot prompting techniques.

Experiment 2: Blog on Applications of AI in Healthcare

Aim:

To design Zero-shot, One-shot, and Few-shot prompts to generate a 200-word blog on “Applications of Artificial Intelligence in Healthcare.”

Procedure:

1. Open Google Colab and create a code cell.
2. Prepare three prompts using different prompting techniques.
3. Include healthcare applications such as diagnosis, medical imaging, and patient monitoring.
4. Display the prompts using Python.
5. Compare the prompts based on their instructions and examples.

Prompt:

zero_shot = """
Write a 200-word blog titled "Applications of Artificial
Intelligence in Healthcare".
Include diagnosis, medical imaging, drug discovery,
patient monitoring, benefits, and challenges.
"""

one_shot = """
Example:

```
Title: AI in Education
Blog: AI improves education through personalized learning
and automated assessment.

Now write a 200-word blog titled
"Applications of Artificial Intelligence in Healthcare".
Follow a similar writing style.
"""

few_shot = """
Example 1:
Topic: AI in Education
Style: Introduction, applications, and conclusion.

Example 2:
Topic: AI in Agriculture
Style: Explain uses, benefits, and challenges.

Now write a 200-word blog on Applications of AI in Healthcare.
Include diagnosis, medical imaging, drug discovery,
patient monitoring, benefits, and limitations.
"""

print("Zero-shot Prompt:\n", zero_shot)
print("\nOne-shot Prompt:\n", one_shot)
print("\nFew-shot Prompt:\n", few_shot)
```

Result:

Three prompts were designed to generate a healthcare blog of approximately 200 words using different prompting techniques.

Experiment 3: Article Summarization into 50 Words

Aim:

To design Zero-shot, One-shot, and Few-shot prompts to summarize an article into 50 words and compare the summaries based on accuracy, completeness, and readability.

Procedure:

1. Open Google Colab and create a code cell.
2. Store the article in a Python string variable.
3. Prepare Zero-shot, One-shot, and Few-shot summarization prompts.

4. Instruct each prompt to produce a 50-word summary.
5. Compare the generated summaries based on accuracy, completeness, and readability.

Prompt:

```
article = """
Artificial Intelligence is transforming healthcare by helping
doctors analyze medical images and identify disease patterns.
AI can monitor patients and support clinical decisions.
It may accelerate drug discovery and reduce administrative work.
However, healthcare AI requires reliable data, privacy protection,
fairness, and human supervision.
"""

zero_shot = f"""
Summarize the following article in exactly 50 words.
Preserve the main ideas without adding new information.

Article: {article}
"""

one_shot = f"""
Example:
Article: Online learning provides flexible education
but requires internet access.

Summary: Online learning offers flexible education
but depends on internet access.

Now summarize this article in exactly 50 words:
{article}
"""

few_shot = f"""
Example 1:
Article: Solar energy produces electricity from sunlight.
Summary: Solar energy converts sunlight into electricity.

Example 2:
Article: Exercise improves fitness and health.
Summary: Exercise supports physical fitness and health.

Summarize the following article in exactly 50 words.
Include the main ideas without adding unsupported facts.
```

```
Article: {article}
"""

print("Zero-shot Prompt:\n", zero_shot)
print("\nOne-shot Prompt:\n", one_shot)
print("\nFew-shot Prompt:\n", few_shot)
```

Result:

Three summarization prompts were designed. Their generated outputs can be evaluated based on accuracy, completeness, readability, and word count.

Experiment 4: Professional Email Requesting Leave Due to Illness

Aim:

To design Zero-shot, One-shot, and Few-shot prompts to generate a professional email requesting leave due to illness.

Procedure:

1. Open Google Colab and create a code cell.
2. Prepare three prompts with different levels of examples.
3. Include the subject, greeting, reason for leave, date, and closing.
4. Display the prompts using Python.
5. Compare the generated emails based on tone, grammar, formatting, and completeness.

Prompt:

```
zero_shot = """
Write a professional email requesting leave due to illness.
Include a subject, greeting, reason, leave date, and closing.
Use polite and grammatically correct English.
"""

one_shot = """
Example:
Subject: Request for Permission
```

```
Dear Sir/Madam,
I request permission to attend an important appointment.
Kindly consider my request.

Regards,
[Your Name]

Now write a professional email requesting leave due to illness.
Include the subject, reason, leave date, and closing.
"""

few_shot = """
Example 1:
Purpose: Meeting request
Tone: Polite and professional.
Format: Subject, greeting, message, and closing.

Example 2:
Purpose: Assignment extension
Tone: Respectful and concise.
Format: Subject, greeting, explanation, and closing.

Now write a professional email requesting illness leave.
Mention [DATE], explain the reason briefly,
request approval, and include a polite closing.
"""

print("Zero-shot Prompt:\n", zero_shot)
print("\nOne-shot Prompt:\n", one_shot)
print("\nFew-shot Prompt:\n", few_shot)
```

Result:

Three prompts were designed to generate professional illness leave emails. The outputs can be compared based on tone, grammar, formatting, and completeness.

Experiment 5: Promotional Social Media Post for an AI Workshop

Aim:

To design Zero-shot, One-shot, and Few-shot prompts to generate a promotional social media post for an AI Workshop.

Procedure:

1. Open Google Colab and create a code cell.
2. Prepare three prompts for promotional content.
3. Include workshop benefits, practical sessions, and registration details.
4. Add instructions for emojis, hashtags, and an attractive headline.
5. Compare the generated posts based on creativity, relevance, and clarity.

Prompt:

```
zero_shot = """
Create an attractive social media post promoting an AI Workshop.
Mention workshop benefits, practical learning, and registration.
Use emojis and relevant hashtags.
"""

one_shot = """
Example:
Python Workshop!
Learn programming through practical activities.
Register now and improve your coding skills!
#Python #Workshop

Now create a similar promotional post for an AI Workshop.
Mention AI applications, practical sessions, and registration.
"""

few_shot = """
Example 1:
Style: Exciting headline, emojis, and hashtags.

Example 2:
Style: Professional language, benefits, and a call to action.

Create a promotional social media post for an AI Workshop.
Include the headline, workshop benefits, hands-on sessions,
date [DATE], venue [VENUE], and registration details.
Use a friendly tone and relevant hashtags.
"""

print("Zero-shot Prompt:\n", zero_shot)
print("\nOne-shot Prompt:\n", one_shot)
print("\nFew-shot Prompt:\n", few_shot)
```

Result:

Three prompts were designed to generate promotional social media content for an AI Workshop using Zero-shot, One-shot, and Few-shot prompting techniques.